Semantic Proto-Role Labeling with NLI

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March 2026

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1 Introduction

In this project we explore an unconventional approach to semantic proto-role labeling to find out whether it holds up to the straight-forward approach and whether there are benefits. Instead of letting a model directly learn and predict SPRL properties as introduced in Dowty's theories, we want to investigate modeling them as entailment relations and address the labeling as a natural language inference task.

The target sentence becomes the premise, and a considered property becomes the hypothesis. Then a neural NLI model can be used to predict if the hypothesis is entailed or not, representing whether the property is applicable or not. For this, we will be collapsing the customary Likert-scale down to a binary state of entailment and contradiction.

A crucial step is the transition of the properties and their core question into hypotheses fit for NLI. The models will be expecting full natural sentences, not just categories, but there is room for experimentation. We try both formal NL entailments as well as more raw interpretations of the properties and hope to find insight on how to sculpt informative entailments. Finally, the NLI property prediction results can be used as features to predict Dowty's proto-agent and proto-patient labels.
Our aim is to achieve at least similar if not better results when predicting semantic proto-role labels when compared to the normal approach and to explore potential benefits that this approach may yield. Our aim is to achieve at least similar if not better results when predicting semantic proto-role labels when compared to the normal approach and to explore potential benefits that this approach may yield.

2 Methodology

2.1 Probing

Our Objective was to assess performance of the pretrained model on our first set of hypotheses templates which we generated using ChatGPT. We used this to construct a baseline for the following fine-tunings.

2.2 Fine-tuning

We then fine-tuned the model on our dataset using our type-aware hypotheses which we generated based on the SPR1 dataset. After the initial fine-tuning we performed a second one after manually changing the hypotheses that did not perform well.

2.3 Evaluation

We evaluated our results by constructing a binary array of gold-standard labels based on the annotations already included in our test data, which we then compared to the predictions of our fine-tuned model. For evaluation, we used accuracy, precision, recall, F1-score, and Cohen's kappa. These metrics were computed after each fine-tuning step to assess which hypotheses performed well and which required further improvement.

3 Models & Datasets

3.1 Models

The model used is a pre-trained RoBERTa NLI transformer model called roberta-large-mnli. It is available for public use on Hugging Face [1]. It classifies a given premise and hypothesis pair into one of three categories; entailment, neutral and contradiction. The neutral label will be disregarded, as we need a binary prediction.

Model Description directly from Hugging Face: "roberta-large-mnli is the RoBERTa large model fine-tuned on the Multi-Genre Natural Language Inference (MNLI) corpus. The model is a pretrained model on English language text using a masked language modeling (MLM) objective."

RoBERTa models focus on Masked Language Modeling and excel at learning token-level contextual representations and are also well-suited for zero-shot classification. They are state-of-the-art classifiers.

The model was used for direct probing and type classification and then later fine-tuned on the upcoming dataset to hopefully improve performance on the task.

3.2 Datasets

To fine-tune the RoBERTa model we use the original Semantic Proto-Role Labeling dataset first introduced in Reisinger et al [2]. This dataset, referred to as SPR1 in later publishments in this field, was created with sentences from the PropBank corpus. These were hand-picked to contain only verbs in veridical contexts, as to avoid logical operations interfering with direct judgements about the verbs. In addition, clausal arguments were also avoided, as it was judged that the questions associated with the dowtian properties proved difficult to apply to these.

Why spr1? As mentioned, this dataset is referenced and involved in later papers [2][3][4][5], and thus accompanied by various statistics across these works, which we could compare our results to. In addition, it has been crafted with the sole purpose of researching Semantic Proto-Role Labeling. For these reasons, we decided to stick to this dataset.

Specifics and Statistics According to Reisinger et al. [2], the dataset comprises 7,045 verb tokens with 11,913 argument spans from 6,808 sentences. It is of the JSON format.

Each entry includes the context sentence/premise under a "sentence" key. The "vp" key contains a string which encodes which tokens of that premise portray the verb, and which tokens portray the argument. These tags help show a model which tokens the hypotheses will target.

The entries also include the 18 considered properties and their annotated label on a Likert scale. The response labels reach from 1-5 and represent how applicable a given property is for the premise, with 1 meaning "very unlikely" and 5 meaning "very likely". Given our NLI approach, we will be collapsing this Likert scale down to a binary state, 1, 2, 3 and 4, 5 becoming "contradiction" and "entailment" respectively.

The dataset is pre-split into test, dev and train sets.

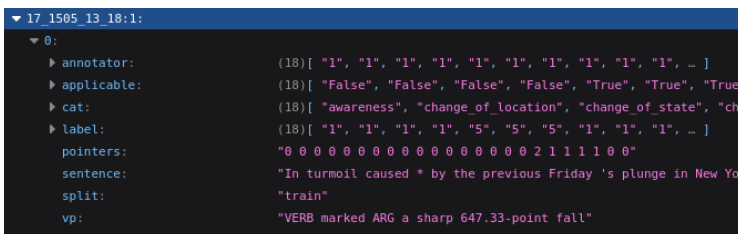

Figure 1: Example entry from the dataset.

3.3 Hypothesis construction

To transform semantic proto-role labeling into an NLI task, we must formulate hypotheses representative of the Dowty-style proto-role properties. These function like features of a sentence which aid in the classification of the sentence as either proto-agent or proto-patient. NLI models are used to natural language sentences as hypotheses; therefore they will be much less likely to succeed when simply using the property names as-is.

This means we need to ideally construct a sentence fitted to and directly referencing the premise while inferring about the core idea of the property. For example, the property "awareness" might look like this: "[ARG] was aware of what was happening as the event occurred."

We experimented with various approaches to generate hypotheses tailored to the sentences in the dataset. Based on the results of probing using these hypotheses and analyzing them with the help of shapey values and other metrics such as the accuracy and F1-measure scores we decided on the best approach to use going forward, which we will get into later.

4 Experiments

4.1 Baseline

As a starting point we established two zero-shot baselines using hypotheses generated by ChatGPT. For the first baseline (*template*), ChatGPT produced a single fixed hypothesis template per proto-role property, yielding one hypothesis per (sentence, argument, property) triple. For the second baseline (*multi-template*), we took the original ChatGPT-generated templates and manually wrote three additional paraphrase variants per property, from which one was sampled at inference time. Both baselines were evaluated by running the stock `roberta-large-mnli` model in zero-shot NLI mode, with no task-specific training. These conditions establish how well off-the-shelf NLI performs on SPR when given generic, argument-agnostic hypotheses.

4.2 Hypothesis Evaluation

To gain insight into which part of the hypothesis drive the model's entailment decision we applied SHAP (SHapley Additive exPlanations) to the stock model. The Shapley values highlight how relevant each token was for the prediction. By analyzing these, we could gain better understanding for what works well and which parts of the hypothesis were obsolete. This helped us decide in which direction to go with hypothesis generation.

Figures 2 and 3 illustrate this process for the *destroyed* property. The original hypothesis "caused a similar coalition to cease to exist" receives a low entailment probability of 0.257, and the SHAP analysis reveals that semantically irrelevant tokens such as *similar* and *coalition* dominate the prediction. After revising the hypothesis to more directly express destruction "a similar coalition was dissolved or ceased to exist as a result of the event", the entailment probability rises to 0.971, with semantically meaningful tokens such as *dissolved*, *ceased*, and *exist* now driving the decision.

Figure 2: SHAP values for original *destroyed* hypothesis (p_entail = 0.257).

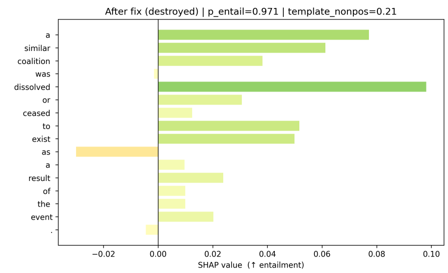

Figure 3: SHAP values after hypothesis revision (p_entail = 0.971).

4.3 Probing

To assess the effect of hypothesis quality independently of fine-tuning, we ran zero-shot probing with three iterations of our type-aware template system, which we had determined to be the most effective hypothesis generation method, each probed using the stock `roberta-large-mnli` model.

In the first iteration (*type-aware v1*), each argument is classified into one of six semantic types: *human*, *organization*, *physical_object*, *location*, *abstract_concept*, or *quantity_or_measure* using zero-shot classification, and a type-specific hypothesis template is selected for each property. Where no type-specific template exists, we fall back to a generic template. Probing results on the development set revealed that several property/type combinations lacked dedicated templates, causing frequent fallback to the generic form and suppressing precision for those properties.

Based on this analysis we extended the template set (*type-aware v2*), adding type-specific entries for the most commonly missing property/type combinations across all six categories, in particular *manipulated_by_another*, *stationary*, *change_of_location*, *change_of_state*, *awareness*, *volition*, *sentient*, and *instigation* for types that previously had no dedicated templates. Zero-shot probing on v2 revealed a further set of high-frequency properties: *destroyed*, *existed_before*, *existed_during*, and *location_of_event*, that were still missing for several types, along with some awareness templates whose wording was not natural enough.

We addressed these remaining gaps in the final iteration (*type-aware v3*) and revised several template wordings for consistency. Zero-shot probing was again run on the v3 pairs to confirm the improvements before fine-tuning.

4.4 Fine-tuning

Fine-tuning was performed twice, each time starting from the stock `roberta-large-mnli` checkpoint. After the first fine-tuning run, several low-frequency properties had fully collapsed, meaning their predictions were entirely skewed toward one class. Continuing training from this checkpoint would have compounded the problem, so the second fine-tuning was restarted from the original stock model.

The first fine-tuning used the type-aware v1 hypothesis pairs as training data. SPR1 Likert labels were binarised using a threshold of 4 (label $\geq 4 \rightarrow$ entailment, label $< 4 \rightarrow$ not_entailment), and rows marked as inapplicable were excluded. The model was trained for 3 epochs with a learning rate of 2×10^{-5}, weight decay of 0.01, and a warmup ratio of 0.06. Per-property F1 on the development set was used to identify which properties had collapsed and which hypothesis templates to improve.

The second fine-tuning followed the same procedure on the type-aware v3 hypothesis pairs, incorporating all template improvements identified through the v1-finetuned and v2 probing analyses. To mitigate the class imbalance observed in the first run, we upsampled positive examples for properties whose positive rate fell below 15% by a factor of 3.

4.5 Evaluation

All conditions were evaluated on both the development and test splits of SPR1. Predictions are binary (*entailment* / *not_entailment*) obtained by thresholding the model's entailment probability at 0.5. We report accuracy, precision, recall, F1-score, and Cohen's κ at both the overall and per-property level. We use per-property F1 throughout as the primary diagnostic for identifying weak hypotheses and tracking the effect of each template revision.

5 Results & Analysis

5.1 Results

Across all experimental runs evaluated on the development set, a clear performance hierarchy emerges that reflects the contribution of each design decision: template strategy, argument type awareness and fine-tuning.

As shown in Figure 4 the three stock model variants perform at a similar level, with F1 scores between 0.524 and 0.541. A common pattern across all of them is the imbalance between precision and recall. In each case recall scores are much higher than precision, which means, that the model tends to predict too many properties, including incorrect ones. As a result, it captures many true cases but also makes many false predictions.

Such behavior is expected for a zero-shot NLI model used on a specialized task. Without task-specific training, the model lacks the ability to confidently reject

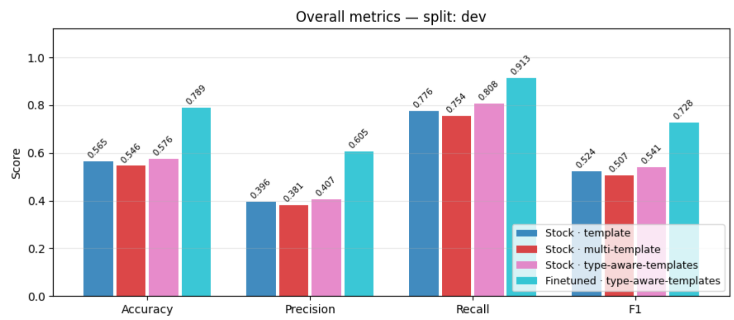

Figure 4: Performance comparison across different model configurations.

incorrect properties, so it tends to over-predict.

Among the stock models, the type-aware templates performs best. This suggests that including information about the argument type helps the model make better predictions. In contrast, the multi-template approach slightly underperforms the single-template baseline (0.507 vs. 0.542). This indicates that simply adding more hypothesis variations does not necessarily improve performance. Without clear guidance, multiple templates can introduce noise instead of improving robustness.

The significant improvement was achieved by fine-tuning. The fine-tuned model reaches a F1 score of 0.728, which is more than 0.18 higher than the best stock model. In addition, repeated fine-tuning plays an important role in performance gains. As shown in Table 1, the fist fine-tuning run already leads to noticeable gains over the stock model, particularly in precision and F1 score.

In the second fine-tuning run we decided to up-sample properties where the positive examples were underrepresented by a factor of 3 to counteract imbalances in the training data.

	N	Accuracy	Precision	Recall	F1	Kappa
Stock · type-aware-templates	19,278	0.5763	0.4066	0.8077	0.5409	0.2204
1st fine-tuning	19,278	0.6346	0.4524	0.8662	0.5943	0.3170
2nd fine-tuning	19,278	**0.7893**	**0.6055**	**0.9134**	**0.7282**	**0.5674**

Table 1: Performance comparison across models on the test set.

Overall, this progression indicates that the model has to learn to better distinguish between relevant and irrelevant properties, leading to more balanced and accurate predictions.

5.2 Proto-Role Prediction

To evaluate whether the predicted proto-role properties capture meaningful information about argument roles, we trained a a Logistic Regression classifier to predict *proto-agent* and *proto-patient* labels based on the property scores produced by our models. As a simple baseline, we use a rule-based classifier that assigns proto-agent or proto-patient labels based on the average scores of their respective proto-role properties.

Using property scores from the **stock model**, the Logistic Regression classifier achieves an **F1** score of **0.787** and an accuracy of 0.855, clearly outperforming the Naive baseline (F1 = 0.575, accuracy = 0.580). This indicates that even stock model can distinguish between proto-agent and proto-patient roles. However, the confusion matrix (Figure 5) reveals that performance is not balanced. The classifier performs much better on proto-patient instances (F1 = 0.91) than on proto-agent (F1 = 0.67). A large number of proto-agent instances are misclassified.

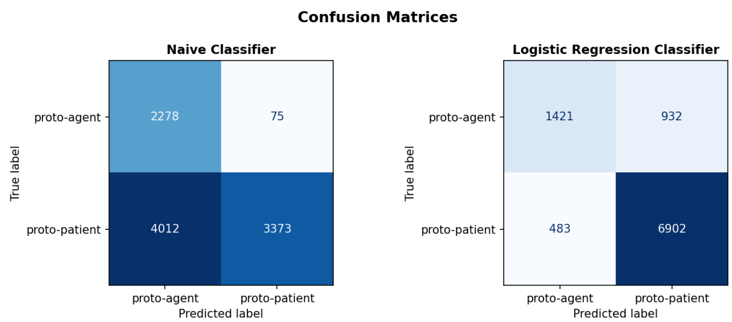

Figure 5: Confusion Matrices of Stock model.

Using property scores from the **fine-tuned** model, performance improves substantially. The classifier reaches an **F1** score of **0.892** and an accuracy of 0.920, while the Naive baseline shows only minor changes. As shown in Figure 6, the confusion matrix becomes much more balanced. In particular, the F1 score for proto-agent increases from 0.67 to 0.84, and the number of misclassified proto-agent instances is greatly reduced. This indicates that the fine-tuned model capture the distinction between agent-like and patient-like arguments better.

5.3 Analysis

Our results show that **fine-tuning** is the most important factor in improving performance. The fine-tuned model clearly outperforms all stock variants, indicating that task-specific supervision is necessary for this task. In contrast, the differences between the stock models are relatively small, suggesting that

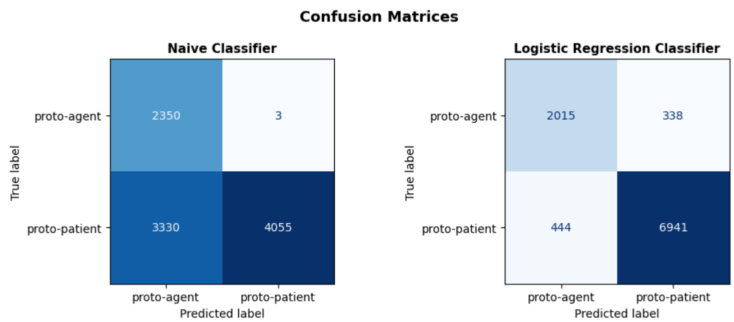

Figure 6: Confusion Matrices of Fine-tuned model.

template design alone has limited impact.

Type-aware templates show a small improvement over the basic template in the stock setting. This indicates that including argument type information in the hypothesis is useful, but not sufficient on its own. However, when combined with fine-tuning, the effect becomes much stronger. Several properties show large improvements, which suggests that type-aware hypotheses provide useful structure that the model can only fully exploit after training.

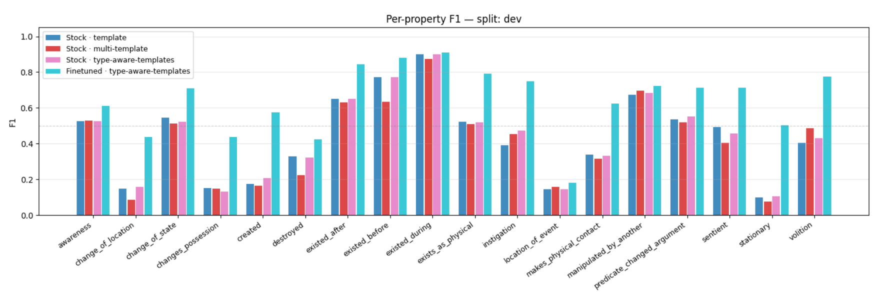

Figure 7: Per-property F1

As shown in Figure 7, performance differs across proto-role properties. Properties related to *time* and *existence* are generally easier to predict and achieve good results across all models. This is likely because they are more directly expressed in the text. In contrast, properties such as *location of event* or *change of possession* remain difficult even after fine-tuning. These cases may require deeper reasoning or additional context.

The results of **proto-role classification** show, that even when using stock model, the classifier is able to distinguish proto-agent from proto-patient roles. As a result we can conclude that proto-role properties already capture useful semantic information. However, the strong improvement with fine-tuning indicates that the quality of these properties is crucial. Better property predictions lead to clearer separation between roles, especially for proto-agent instances, which are harder to identify.

6 Summary & Conclusion

In this work, we explored an unconventional approach to Semantic Proto-Role Labeling by treating property predictions as a Natural Language Inference task. Rather than training a classifier directly on SPR labels, we fine-tuned a RoBERTa-large-MNLI model using carefully crafted, type-aware hypothesis templates derived from Dowty's proto-role properties. Our results demonstrate that this NLI-based formulation is a viable and competitive approach for SPRL.

Property prediction. Our fine-tuned model achieves a micro-F1 of 0.729 and a macro-F1 of 0.638 on the SPR1 test set, with an accuracy of 0.789 and Cohen's $\kappa = 0.567$. Compared to Rudinger et al. [4], the most related prior work, who report a micro-F1 of 0.822 and macro-F1 of 0.693 using a Neural-Davidsonian architecture trained directly on SPR labels, our approach trails on both metrics. However, this gap should be interpreted in context: our model repurposes an off-the-shelf NLI model without a purpose-built SPRL architecture, and the performance difference narrows for harder, lower-frequency properties as evidenced by the smaller macro gap (5.5 points) compared to the micro gap (9.3 points).

Proto-role classification. Using our predicted property scores as features for a Logistic Regression classifier, we achieve an F1 of 0.892 and accuracy of 0.920 for proto-agent/proto-patient classification, a substantial margin over the naïve score-averaging baseline (F1 = 0.575). This confirms that the property predictions produced by our NLI model carry rich semantic information consistent with Dowty's theoretical framework.

Design insights. The iterative refinement of type-aware templates across three probing rounds proved essential. Zero-shot stock models plateau around F1 0.524–0.541 regardless of template strategy, underscoring that template design alone cannot substitute for task-specific supervision. Fine-tuning provides the dominant boost (+0.18 F1), while type-aware hypotheses provide structured inductive bias that the model can fully exploit only after training.

Limitations and future work. Several properties, notably *location of event* and *change of possession*, remain difficult even after fine-tuning, likely requiring broader contextual reasoning beyond what a single sentence provides. Due to property collapse observed in the first fine-tuning run, the subsequent fine-tuning step had to be restarted from the original stock checkpoint rather than continued from the previous one; resolving this instability, for instance through better upsampling strategies, could enable iterative fine-tuning and push performance further. Extending the approach to SPR2 [3], which covers a broader range of predicates, is another natural next step. We also explored using larger generative models for hypothesis generation, though evaluation was limited by strict API usage constraints and did not allow for large-scale testing, preliminary results were mixed and the viability of this direction remains an open question. Nevertheless, this work demonstrates that recasting SPRL as NLI is a principled and effective approach that achieves results competitive with dedicated neural models while offering greater interpretability through its explicit hypothesis structure.

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